

Fracturing? The Productivity Slowdown in the U.S. Manufacturing Sector

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Abstract

This paper examines whether the post-2007 slowdown in U.S. manufacturing productivity was concentrated in a few sectors or broad-based. Using six-digit NAICS data, I compare pre- and post-2007 TFP growth, aggregate sector contributions using Domar weights, and estimate level shift regressions. Roughly 60% of sectors experienced negative shifts in productivity growth, while the top 25% of four-digit sectors account for about 40% of the decline, a share that is not statistically unusual. Emissions-adjusted TFP shows a similar pattern, ruling out productivity gains from emissions reduction. Overall, the evidence indicates a widespread, rather than sector-specific, productivity slowdown. I derive a theoretical model that explains the slowdown through competition with a high-innovation sector for high-skilled labor and investment in product differentiation.

1 Introduction

Motivation. Long-run economic growth is driven primarily by sustained improvements in productivity rather than additional factor inputs, a fact recognized early in the economic literature (Schumpeter, 1942; Solow, 1956). Therefore, having reliable data regarding the trends in productivity is central for policymakers to determine the development path of the economy and the possible instruments needed to support it. Regarding the manufacturing sector, although its share of GDP has gradually fallen over recent decades, it remains a significant component of the U.S. economy, accounting for roughly 10% of total GDP (U.S. Bureau of Economic Analysis, 2025). The decline in productivity growth is particularly striking, given that around two-thirds of corporate R&D spending stems from the manufacturing sector (Autor et al., 2020).

Related Literature. This paper builds on the existing literature of productivity growth and its decline in recent decades. Over the last years, different authors stressed different reasons for the observed decline. For example, Byrne et al. (2013) and Fernald (2014) argue that the slowdown is not connected to the Great Financial Crisis, and mainly stems from a decline in IT or IT-related sectors. Furthermore, while the literature focusing on productivity in the manufacturing sector has been large at the end of last century (see e.g. Hulten and Schwab (1984); Morrison (1997); Bartelsman and Dhrymes (1998)), it is now relatively limited. First thorough insights into the decline in manufacturing productivity stem from Syverson (2016), who shows that labor productivity growth fell sharply from roughly 2.8% per year in 1995–2004 to about 1.3% per year in 2005–2015. Similar trends regarding labor productivity were documented by Lashkari and Pearce (2024). Furthermore, they show that TFP follows the same pattern, although lower in magnitude and that both high- and low-growth sectors experience this decline. Recently, Atalay et al. (2025) argued that there is not a productivity decrease at all, but that the deflators underestimate the true effects of quality improvements.

Still, the literature remains thin as researchers usually focus on the aggregate evolution of productivity and on emerging sectors and technologies instead of the manufacturing sector. My paper aims to counter this lack of granular research by (i) looking at six-digit manufacturing sector-level data of productivity and value added over the last 60 years, provided by the NBER-CES Manufacturing Industry database (Becker et al., 2021) and (ii) providing a formal explanation. The micro-level characteristics are especially important for policy makers when trying to establish how to address this stagnation (i.e. whether targeting a few sectors will increase aggregate productivity or if broad-based support is needed). I find that around 60% of six-digit sectors experienced a decline in the growth rate of TFP and that the aggregate decline cannot be explained by a few sectors.

Outline. The rest of the paper is structured as follows: Section 2 presents the data and methodology, Section 3 discusses the results. Section 4 provides a framework that can explain the results and Section 5 concludes.

2 Empirical Strategy and Data

The primary dataset is the annual U.S. manufacturing panel from the NBER-CES Manufacturing Industry Database, covering 1958–2018 at the six-digit NAICS level. For each industry i and year t , the dataset provides, among others, five-factor productivity (`tfp5`), its annual growth rate (`dtfp5`), and value added (`vadd`). Five-factor productivity is used in order to have a more precise measurement of productivity as it includes, apart from labor and capital, other production factors. Namely energy, materials, and purchased services. For robustness checks, I complement these data with aggregate manufacturing TFP from the U.S. Bureau of Labor Statistics (2025) (`MFGPROD`), manufacturing CO₂ emissions from the U.S. Environmental Protection Agency (2024) (`Gross_total`) and a recent, high estimate of the social cost of carbon (SCC) from Bilal and Känzig (2024).

2.1 Baseline Measurement of the Productivity Slowdown

The central object of interest is the shift in trend TFP growth around 2007, the year in which U.S. (manufacturing) productivity growth began to slow. I begin by splitting the sample into a pre-2007 period and a post-2007 period. For each six-digit industry i , I compute the geometric average annual TFP growth rate in each subperiod, denoted g_i^{pre} and g_i^{post} . The sector-specific slowdown is then defined as:

$$\Delta g_i \equiv g_i^{\text{post}} - g_i^{\text{pre}}$$

A negative value of Δg_i indicates that industry i experienced slower average TFP growth after 2007, therefore providing a measure to analyze long-term evolution. The distribution of Δg_i provides a transparent way to assess how widespread the slowdown was across manufacturing industries, independent of their economic size.

2.2 Aggregation and Sector Contribution Analysis

To determine whether the aggregate slowdown is driven by a small number of large sectors or reflects a broad-based decline, I next analyze contributions at the four-digit sector level. It could be the case that just a large number of very small sectors experienced a decline, while the aggregate decline was driven by just a few sectors. Aggregation to the four-digit level (denoted j) is helpful because the number of six-digit (denoted i) industries would otherwise make the top contributors share mechanically small and thus hard to interpret. Statistical rigor will prevent the definition of an arbitrary cutoff after which I can conclude that the share is disproportionate. However, I will run a bootstrapping test to see whether the observed concentration strongly differs from what would arise if sectoral slowdowns were equally assigned across sectors. For each sector i and j , I compute the following value-added share, using averages over time:

$$w_{j,i} = \frac{\text{value added of sector } j \text{ or } i}{\text{total manufacturing value added}}$$

Which serves as approximation to Domar weights.¹ The contribution of sector j to the slowdown is defined as:

$$c_j \equiv w_j \Delta g_j$$

Following Hulten's theorem (Hulten, 1978), the change in TFP growth at the four-digit level can be approximated by the weighted sum of six-digit slowdowns:

$$\Delta g_j \approx \sum_{i \in j} w_i \Delta g_i$$

Thus, $w_i \Delta g_i$ represents the contribution of six-digit industry i to the slowdown within its parent four-digit sector j . I focus on the top five four-digit sectors to evaluate whether they account for a disproportionate share of the aggregate decline.

2.3 Robustness Checks

As a robustness exercise, I estimate sector-specific time-series regressions that allow for a level shift in 2007. For each six-digit industry i , I estimate the following model:

$$\Delta TFP_t = \alpha + \beta \cdot t + \delta \cdot \mathbf{1}\{t \geq 2007\} + \varepsilon_t$$

Where β captures the average trend in TFP growth and δ measures the level shift beginning in 2007. A negative coefficient δ_i indicates that industry i experienced a decline in its trend growth rate. The distribution of δ_i is then plotted similar to Δg_i for comparison.

As a second robustness check, I examine whether adjusting TFP for emissions alters the interpretation of the post-2007 slowdown in the general setting (i.e. checking whether there was even a decrease in productivity). This is done to rule out possible productivity increases via emissions reduction. Following the proposition of emissions-adjusted TFP (TFPE) as the relevant welfare measure by De Ridder and Rachel (2025), I construct it as follows:

$$TFPE_t \equiv TFP_t \left(1 - \frac{E_t}{Y_t} SCC_t \right)$$

¹Value-added instead of gross output is used to avoid double counting of intermediate inputs.

Where E_t denotes manufacturing emissions (in metric tons), Y_t denotes value added, and SCC_t is the social cost of carbon, taken as \$928 in 2017 USD from Bilal and Känzig (2024).² This measure captures productivity gains associated with emissions reductions, which conventional TFP does not record. I compare the evolution of TFP and TFPE to assess whether emissions-adjusted productivity displays a similar post-2007 slowdown.

3 Results

3.1 Baseline Evidence

I begin by examining the change in average TFP growth across six-digit manufacturing industries. Figure 1 plots the distribution of Δg_i , defined as the difference between post-2007 and pre-2007 average TFP growth. 68% of sectors experienced a negative shift, indicating that the productivity slowdown was widespread rather than confined to a few industries. Additionally, the plot reveals approximately a normal distribution, centered around a value slightly less than zero with values between -0.1 and 0.1. Since the yearly changes in TFP are mostly values between ± 0.03 , this variance could hint on some dominant sectors. In the next section, I formally examine this possibility.

3.2 Aggregation and Sector Contributions

To assess whether the aggregate slowdown is disproportionately driven by a few large sectors, I analyze the previously calculated contributions. Figure 2 shows the contributions of the top five four-digit sectors. From a total of 20 four-digit NAICS manufacturing sectors, the top 5 explain about 40% of the aggregate slowdown. The bootstrapping test for significance results in a p-value of 0.26, indicating that the observed contribution of the top 25% of sectors is not statistically distinguishable from what would arise under equal shares.

²I chose a recent, high estimate of the social cost of carbon to ensure maximal robustness.

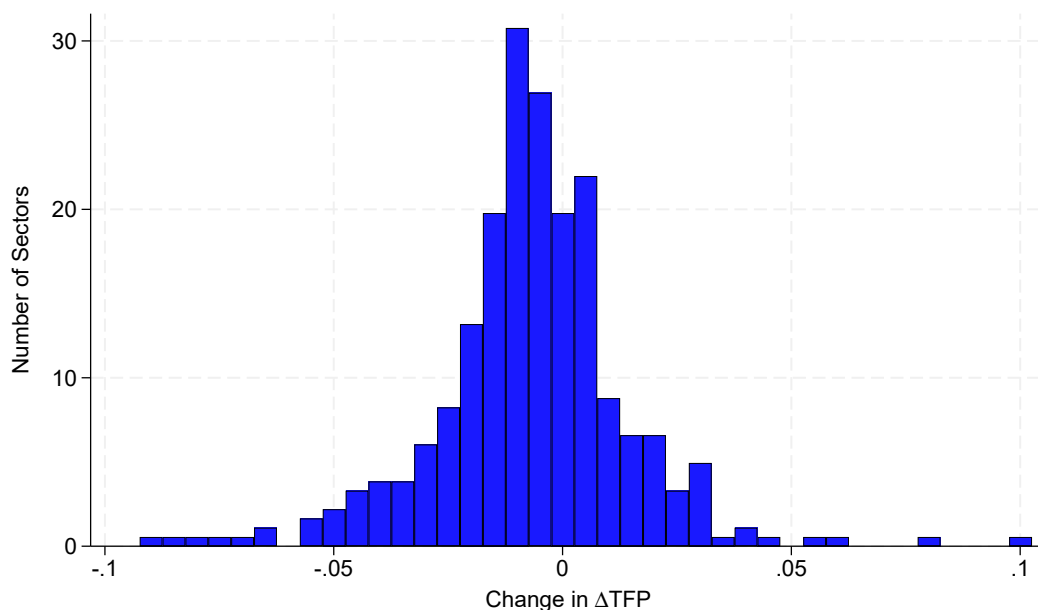


Figure 1: Distribution of Sector-Level Changes in TFP growth $\{\Delta g\}$. Source: CES.

This finding therefore rules out explanations that attribute the decline primarily to a narrow subset of large (“superstar”) manufacturing industries. Instead, it suggests that the aggregate slowdown reflects declines across many sectors of economically meaningful size, consistent with a common, economy-wide mechanism.

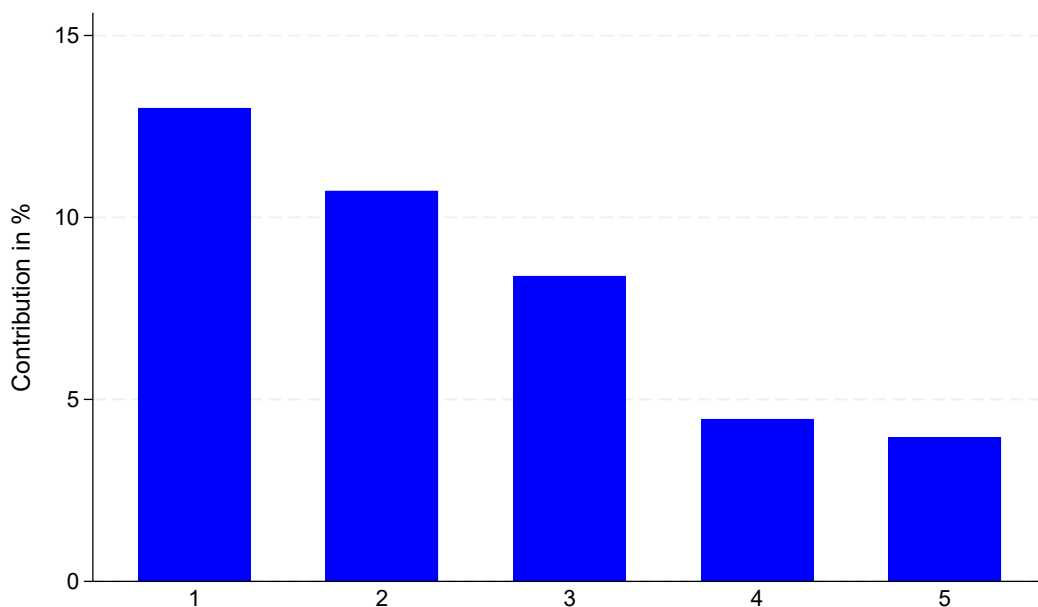


Figure 2: Contribution of Top 5 four-digit Sectors to Aggregate Slowdown. Source: CES.

3.3 Robustness Evidence

As a first robustness check, I examine whether the slowdown reflects a persistent shift in TFP growth, using the sector-specific level shift regression. Figure 3 displays the distribution of these coefficients. Similar to the pre/post comparison, most δ_i values are negative (58%), confirming that the slowdown is reflected in long-run, broad-based trends rather than short-term fluctuations. The distribution matches the previous findings as well. Most estimates are again clustered around zero, with two outliers at -0.2. The correlation between Δg_i and δ_i is 0.81, indicating that they measure the same thing but with a different approach.

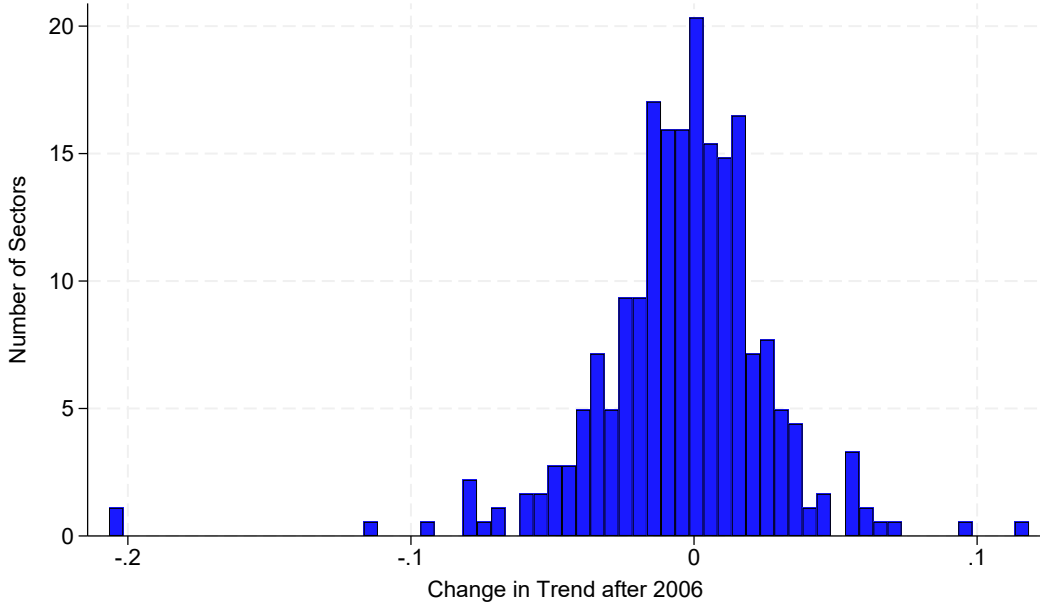


Figure 3: Distribution of Estimated Level Shifts $\{\delta_i\}$ in Sector-Level TFP. Source: CES.

The second robustness check uses emissions-adjusted TFP (TFPE). Figure 4 plots the evolution of aggregate TFP alongside TFPE. The similarity of the two series after 2007 strengthens the main finding: Productivity growth stagnated even after accounting for the possibility of productivity increases by emissions reduction. It is worth noting that, before 2007, TFPE grew more rapidly than TFP. This implies that the manufacturing sector was able to cut emissions while simultaneously raising its productivity, suggesting that these two objectives do not necessarily conflict. However, that also depends on the SCC estimate used.

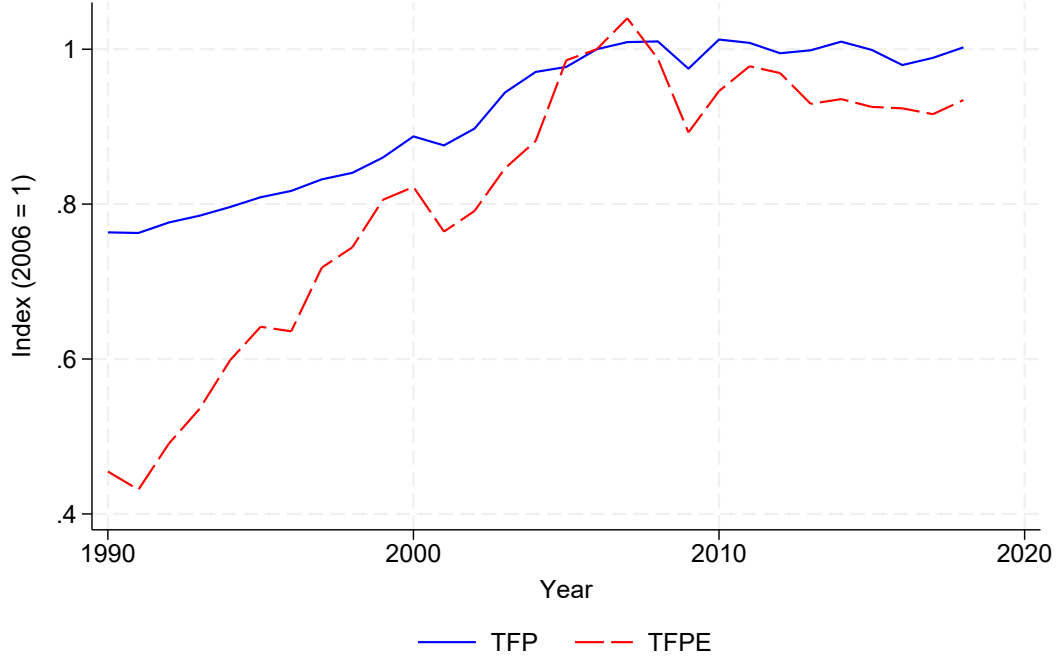


Figure 4: Evolution of TFP and TFPE. Source: CES and EPA.

4 Theoretical Explanation

This section, motivated by the recent remarks of Atalay et al. (2025) that quality improvements are not adequately reflected in TFP measurements, develops a model that offers a potential explanation for the broad-based decline in observed TFP. To be consistent with other trends observed in recent years, namely rising markups and higher R&D expenditures, the model incorporates these trends as well and presents a unified explanation. The key intuition behind this model is that in recent years, high innovation sectors emerged which attract skilled workers with high wages, e.g. Silicon Valley companies. Furthermore, manufacturing firms increase their profit share, possibly by product differentiation and quality improvements, which I assume requires some sort of R&D expenditure. Because these quality improvements are reflected in higher markups and not higher productivity, it is not adequately reflected in TFP.

4.1 Environment

Competition and Labor Supply: Time is continuous. A high-innovation sector D takes as input high-skill labor and produces output according to:

$$Y_t^D = Z_t(L_t)^\beta, \quad \beta \in (0, 1) \quad (1)$$

Where productivity Z_t is an exogenously given increasing process. Firms choose labor to maximize profits, implying the labor demand condition:

$$L_t^D = \left(\frac{\beta Z_t}{w_t^H} \right)^{\frac{1}{1-\beta}} \quad (2)$$

An increase in productivity, Z_t , raises labor demand at any given wage. Therefore, the wage w_t^H must increase as well to match the demand. High-skill labor is supplied according to the following function:

$$L_t = \kappa \frac{2}{\pi} \times \arctan(w_t^H) \quad (3)$$

Which is increasing and concave in the argument, and bounded above by κ . I assume a reduced-form, but a micro-founded household block could be incorporated and would not change the qualitative predictions.

Manufacturing: The sector features a mass one of lines (firms). Lines are heterogeneous in technology and market power and produce a good that has some degree of differentiation. Each manufacturing line ℓ produces its final good using low-skill labor:

$$y_\ell = a_\ell n_\ell \quad (4)$$

Where n_ℓ is low-skill labor hired at wage w_t^L and a_ℓ is productivity. Optimal pricing implies a constant markup over marginal cost:

$$\mu_\ell \equiv \frac{p_\ell}{c_\ell} = \frac{\varepsilon_\ell}{\varepsilon_\ell - 1} \quad (5)$$

I assume $\varepsilon > 2$, such that markups do not exceed 100%, which matches empirical findings. Each line faces iso-elastic residual demand with a line-specific elasticity:

$$q_\ell = B_t p_\ell^{-\varepsilon_\ell} \quad (6)$$

Where B_t is a common demand shifter determined by aggregate expenditure. This yields revenue $p_\ell q_\ell = B_t p_\ell^{1-\varepsilon_\ell}$, and the revenue share of line ℓ is:

$$s_\ell \equiv \frac{p_\ell q_\ell}{\int_0^1 p_k q_k dk} \quad (7)$$

Using optimal pricing, the revenue share can be written in terms of primitives:

$$s_\ell = \frac{B_t \left(\frac{\mu_\ell w_t^L}{a_\ell} \right)^{1-\varepsilon_\ell}}{\int_0^1 B_t \left(\frac{\mu_k w_t^L}{a_k} \right)^{1-\varepsilon_k} dk} \quad (8)$$

This formulation is consistent with, but does not require, a specific utility function.³ Firms can allocate their resources between two types of R&D. *Vertical* innovation R_v , which reduces marginal cost c_ℓ through higher efficiency (TFP), while *horizontal* innovation R_h increases market power by lowering the effective elasticity ε_ℓ through differentiation, thereby increasing μ_ℓ . The processes for the two types of innovation are assumed to be independent and measured such that one unit of R uses one unit of labor. Additionally, I assume that firms face fixed costs for both types of research, such as lab equipment and machines.

Vertical innovation: Arrival in a line is Poisson with intensity:

$$\lambda_v(R_v; X_t) = \alpha_v \frac{(R_v - \psi_v)^{1-\eta}}{1 + \phi X_t}, \quad \eta \in (0, 1), \alpha_v, \phi > 0, R_v \geq \psi_v \quad (9)$$

This captures diminishing returns in R&D ($\eta \in (0, 1)$), fixed costs ψ_v and a fishing-out effect $(1 + \phi X_t)$, following the proposition of Bloom et al. (2020) that ideas are

³A nested CES aggregator as in Atkeson and Burstein (2010) would micro-found this demand structure.

getting harder to find. X_t represents the manufacturing technological frontier given by $X_t \equiv \sup\{a_{\ell,t}\}$. A successful vertical innovation increases efficiency a_ℓ multiplicative by a factor $\lambda_a > 1$.

Horizontal innovation: Arrival in a line is Poisson with intensity:

$$\lambda_h(R_h) = \alpha_h(R_h - \psi_h)^{1-\nu}, \quad \nu \in (0, 1), \quad \alpha_h > 0, \quad R_h \geq \psi_h \quad (10)$$

Which features diminishing returns and fixed costs but no fishing-out in differentiation. On success, the firm increases market power in its line with $\varepsilon^+ = 1 + \lambda_\varepsilon(\varepsilon - 1)$ and $\lambda_\varepsilon < 1$. Taking the three labor demand conditions, market clearing implies:

$$L_t = L^D(w_t^H, t) + \int_0^1 R_{v,\ell}(w_t^H, t) + R_{h,\ell}(w_t^H, t) d\ell \quad (11)$$

Therefore, in equilibrium, the rise in the high-innovation sectors productivity will increase its labor demand, thereby pushing w_t^H upwards. Since manufacturing R&D uses the same high-skill labor and pays the same wage, it will also be affected.

4.2 Firms Problem

Let the state of a line be $s = (a, \varepsilon)$ (equivalently (a, μ)). Let $V(a, \varepsilon, t)$ denote the value of being the firm in a line at time t and $\pi(\mu(\varepsilon), a)$ denote the flow profit. The firm takes wages as given and chooses (R_v, R_h) to maximize expected present value. Let $r > 0$ be the discount rate. The Hamilton–Jacobi–Bellman (HJB) equation is:⁴

$$\begin{aligned} rV(a, \varepsilon, t) = & \pi(\mu(\varepsilon), a) - w_t^H(R_v + R_h) + \partial_t V(a, \varepsilon, t) \\ & + \lambda_v(R_v; X_t)[V(a^+, \varepsilon, t) - V(a, \varepsilon, t)] \\ & + \lambda_h(R_h)[V(a, \varepsilon^+, t) - V(a, \varepsilon, t)] \end{aligned} \quad (12)$$

Because λ_v and λ_h are concave in R , the optimal policy features interior choices. Let $\Delta_v(a, \varepsilon, t) \equiv V(\lambda_a a, \varepsilon, t) - V(a, \varepsilon, t)$ and $\Delta_h(a, \varepsilon, t) \equiv V(a, \varepsilon^+, t) - V(a, \varepsilon, t)$. The first-order conditions for an interior optimum are:

$$w_t^H = \frac{\partial \lambda_v}{\partial R_v}(R_v, X_t) \Delta_v = \alpha_v \frac{1 - \eta}{1 + \phi X_t} (R_v - \psi_v)^{-\eta} \Delta_v \quad (13)$$

$$w_t^H = \frac{\partial \lambda_h}{\partial R_h}(R_h) \Delta_h = \alpha_h (1 - \nu) (R_h - \psi_h)^{-\nu} \Delta_h \quad (14)$$

Rearranging yields policy rules:

$$R_v(a, \varepsilon, t) = \psi_v + \left(\alpha_v \frac{1 - \eta}{w_t^H (1 + \phi X_t)} \Delta_v(a, \varepsilon, t) \right)^{1/\eta} \quad (15)$$

$$R_h(a, \varepsilon, t) = \psi_h + \left(\alpha_h \frac{1 - \nu}{w_t^H} \Delta_h(a, \varepsilon, t) \right)^{1/\nu} \quad (16)$$

4.3 Model Takeaways

Broad-based productivity growth: TFP growth is tied to the evolution of output-weighted a_ℓ . Since only vertical innovation raises a , a decline in $\lambda_v(R_v, X_t)$ at the line level slows TFP growth. Proposition 1 in the appendix shows that vertical innovation intensity λ_v^* is unambiguously decreasing in the high-skill wage w_t^H , even after accounting

⁴The derivation can be found in the appendix.

for the endogenous response of the value gain Δ_v . The argument does not require holding Δ_v fixed, as the envelope theorem implies $\partial V/\partial w^H < 0$, and convexity of V in a ensures that higher-productivity firms lose more value when wages rise, so $\partial \Delta_v/\partial w^H \leq 0$. This reinforces the direct cost effect, making the total response strictly negative. Furthermore, if only a minority of sectors experience a decline, the reallocation effect would offset this negative trend because these sectors' output shares decline. This means that, given an aggregate decline, the model predicts an underlying broad-based decline.

Furthermore, if only a minority of sectors experience a decline, the reallocation-effect would offset this negative trend because these sector's output shares decline. This means that, given an aggregate decline, the model predicts an underlying broad-based decline.

Rising output-weighted markups: Aggregate markups rise for two complementary reasons. First, within-line effects: Horizontal innovations directly increase a product line's markup when they arrive. Second, reallocation effects: Vertical innovations and selection shift market shares toward more productive, higher-markup lines, while low-productivity, low-markup lines lose shares.

Rising R&D expenditure with slowing TFP: Even as vertical innovation becomes harder and its productivity payoff declines, aggregate R&D expenditure rises in equilibrium. Each manufacturing firm must employ a fixed amount of high-skill labor to operate its R&D activities. As productivity growth in the high-innovation sector raises demand for high-skill labor, the equilibrium wage w_t^H increases. The fixed R&D labor requirement therefore translates rising wages directly into higher aggregate R&D spending, even as the intensity of vertical innovation and measured TFP growth decline.⁵

⁵A formal derivation of the model takeaways can be found in the appendix.

5 Conclusion

This paper examined whether the post-2007 slowdown in U.S. manufacturing productivity growth was driven by a small set of dominant sectors or reflects a broad-based decline. The empirical evidence supports the latter interpretation. Over 60% of six-digit industries experienced negative shifts in average TFP growth starting in 2007, and while the top five four-digit sectors accounted for about 40% of the aggregate slowdown, a bootstrap analysis showed that this concentration is not statistically unusual. Level shift regressions reinforced this conclusion: Most sectors exhibit persistent declines in trend productivity growth. Emissions-adjusted TFP moved closely with conventional TFP, demonstrating that the slowdown persists even when accounting for productivity gains induced by reduced emissions.

The theoretical framework developed in this paper provides a unified mechanism consistent with the observed combination of a broad-based decrease in productivity growth, rising markups, and increasing R&D expenditures. In the model, sustained expansion of a high-innovation frontier sector raises the equilibrium wage of high-skill labor, increasing the cost of innovation economy-wide. As innovation becomes more expensive and vertical innovation faces increasing fishing-out effects, firms gradually reallocate innovative effort away from productivity growth toward market-power enhancing activities, leading to a persistent slowdown in aggregate measured productivity. Importantly, the resulting stagnation reflects not sector-specific failure, but an equilibrium response to economy-wide competition for high-skill workers.

Nevertheless, several limitations remain. The empirical analysis is descriptive and does not identify causes, and six-digit sectors still represent aggregation of different firms. The theoretical model builds on vague assumptions regarding the path of wages. Future research could therefore refine the mechanisms, incorporate this into a full model of creative destruction, and empirically test the model's prediction regarding the relationship between firm markups and innovation persistence.

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A Appendix

A.1 Derivation of the HJB

Fix a line in state (a, ε) at time t . Over dt , the line earns $\pi(\mu(\varepsilon), a) dt$, pays $w_t^H(R_v + R_h) dt$, and transitions according to independent Poisson shocks:

- Vertical success with probability $\lambda_v(R_v, X_t) dt$: $(a, \varepsilon) \rightarrow (\lambda_a a, \varepsilon)$
- Horizontal success with probability $\lambda_h(R_h) dt$: $(a, \varepsilon) \rightarrow (a, \varepsilon^+)$

The firm's problem is then given by:

$$V(a, \varepsilon, t) = \max_{R_v, R_h \geq \psi} \left\{ [\pi(\mu(\varepsilon), a) - w_t^H(R_v + R_h)] dt + e^{-r dt} \mathbb{E}[V(a', \varepsilon', t + dt) \mid a, \varepsilon] \right\}$$

Using the transition structure and expanding to first order gives:

$$\begin{aligned} \mathbb{E}[V(a', \varepsilon', t + dt)] &= V(a, \varepsilon, t + dt) + \lambda_v(R_v, X_t) dt [V(\lambda_a a, \varepsilon, t + dt) - V(a, \varepsilon, t + dt)] \\ &\quad + \lambda_h(R_h) dt [V(a, \varepsilon^+, t + dt) - V(a, \varepsilon, t + dt)] \end{aligned}$$

Using a first order Taylor Approximation $e^{-r dt} \approx 1 - r dt$, $V(a, \varepsilon, t + dt) = V(a, \varepsilon, t) + \partial_t V(a, \varepsilon, t) dt$ and dividing by dt yields the Hamilton–Jacobi–Bellman equation (12).

A.2 Decline in Productivity Growth

Aggregate manufacturing productivity is defined as the revenue-weighted average of line-level productivities:

$$A_t \equiv \int_0^1 s_{\ell t} a_{\ell t} d\ell \tag{17}$$

Where $s_{\ell t}$ is defined by (8).⁶ Where s_{ℓ} is firm ℓ 's output share. Because Poisson arrival is in jumps, the instantaneous expected drift is given by differentiating A_t , applying the

⁶This is an approximation to the Domar-weighted aggregate. With heterogeneous markups, the exact Domar decomposition involves additional terms; see Hulten (1978).

product rule, taking expectations and plugging in the law of motion for $a_{\ell t}$.

$$\frac{d}{dt} \mathbb{E}[A_t] = \underbrace{\int_0^1 \mathbb{E}[a_{\ell t} \dot{s}_{\ell t}] d\ell}_{\text{reallocation}} + \underbrace{(\lambda_a - 1) \int_0^1 \mathbb{E}[s_{\ell t} a_{\ell t} \lambda_{v,\ell t}] d\ell}_{\text{within-line vertical innovation}} \quad (18)$$

The first term captures reallocation of revenue shares across lines, while the second term captures the direct contribution of vertical innovation to aggregate productivity. Under the optimal R&D policy, vertical innovation intensity is:

$$\lambda_{v,\ell t}^* = \alpha_v^{1/\eta} (1 - \eta)^{\frac{1-\eta}{\eta}} \frac{\Delta_{v,\ell t}^{\frac{1-\eta}{\eta}}}{w_t^H \frac{1-\eta}{\eta} (1 + \phi X_t)^{1/\eta}} \quad (19)$$

Holding $\Delta_{v,\ell t}$ fixed, the comparative statics are:

$$\frac{\partial \log \lambda_{v,\ell t}^*}{\partial \log w_t^H} = -\frac{1 - \eta}{\eta} < 0, \quad \frac{\partial \log \lambda_{v,\ell t}^*}{\partial \log (1 + \phi X_t)} = -\frac{1}{\eta} < 0 \quad (20)$$

Thus, increases in the high-skill wage and in the frontier level X_t reduce the intensity of vertical innovation at the line-level. Provided the reallocation term does not increase sufficiently to offset these forces, the expected drift of aggregate productivity growth declines over time.

A.3 Total Effect of Wages on Innovation Intensity

Proposition 1. *Under the assumption that $\varepsilon > 2$, vertical innovation intensity $\lambda_{v,\ell}^*$ is strictly decreasing in the high-skill wage w_t^H , accounting for the endogenous response of $\Delta_v(a, \varepsilon, t) \equiv V(\lambda_a a, \varepsilon, t) - V(a, \varepsilon, t)$.*

Proof. The proof proceeds in three steps.

Step 1: V is decreasing in w^H . By the envelope theorem applied to the HJB (12), at the optimal policy (R_v^*, R_h^*) :

$$\frac{\partial V(a, \varepsilon, t)}{\partial w_t^H} = -(R_v^*(a, \varepsilon, t) + R_h^*(a, \varepsilon, t)) < 0 \quad (21)$$

since both R&D inputs are strictly positive at an interior optimum.

Step 2: V is convex in a . Define the cone $\mathcal{C} = \{V : V \text{ is convex in } a\}$. Under assumption, flow profits $\pi(\mu(\varepsilon), a) \propto a^{\varepsilon-1}$ are convex in a since $\varepsilon - 1 > 1$. Given any $V \in \mathcal{C}$, the value gain $\Delta_v(a) = V(\lambda_a a, \varepsilon, t) - V(a, \varepsilon, t)$ is increasing in a (a property of convex functions evaluated at a multiplicative shift with $\lambda_a > 1$). The optimal R&D policies $R_v^*(a), R_h^*(a)$ are increasing in a through their dependence on Δ_v and Δ_h , and the Bellman operator maps $\mathcal{C}$ into itself. Since the operator is a contraction, the unique fixed point $V^* \in \mathcal{C}$ by Banach's Fixed Point Theorem.

Step 3: $\partial \Delta_v / \partial w^H \leq 0$. From Steps 1 and 2:

$$\frac{\partial \Delta_v}{\partial w^H} = \frac{\partial V(\lambda_a a, \varepsilon)}{\partial w^H} - \frac{\partial V(a, \varepsilon)}{\partial w^H} = -[(R_v^* + R_h^*)|_{\lambda_a a} - (R_v^* + R_h^*)|_a] \leq 0 \quad (22)$$

where the inequality follows because $R_v^* + R_h^*$ is increasing in a (Step 2). Define $\Omega_v \equiv \Delta_v / w_t^H$. Then:

$$\frac{d\Omega_v}{dw^H} = \frac{1}{w^H} \frac{\partial \Delta_v}{\partial w^H} - \frac{\Delta_v}{(w^H)^2} < 0 \quad (23)$$

Since λ_v^* is strictly increasing in $\Omega_v / (1 + \phi X_t)$ by (9), the result follows. $\square$

A.4 Rising Aggregate Markups

Define the output-weighted aggregate markup as:

$$M_t \equiv \int_0^1 s_\ell(t) \mu_\ell(t) d\ell \quad (24)$$

Again, applying the product rule and taking expectations yields the decomposition:

$$\frac{d}{dt} \mathbb{E}[M_t] = \underbrace{\int_0^1 \mathbb{E}[\mu_\ell(t) \dot{s}_\ell(t)] d\ell}_{\text{reallocation}} + \underbrace{\int_0^1 \mathbb{E}[s_\ell(t) \lambda_h(R_{h,\ell}(t)) (\mu(\varepsilon_\ell^+) - \mu(\varepsilon_\ell))] d\ell}_{\text{within-line}} \quad (25)$$

Horizontal innovation contributes directly to the within-line term by raising the line's markup $\mu(\varepsilon_\ell^+) > \mu(\varepsilon_\ell)$. Vertical innovation contributes through the reallocation term by increasing productivity a_ℓ , therefore and raising revenues, which increases the revenue

share s_ℓ of successful lines. As a result, high-markup lines tend to gain share over time, so that $\text{Cov}(\mu_\ell, \dot{s}_\ell) > 0$ along the transition, pushing aggregate markups M_t upward.

A.5 Rising Aggregate R&D Expenditure

Aggregate manufacturing R&D expenditure is given by:

$$\mathcal{E}_t = w_t^H \int_0^1 (R_{v,\ell}(t) + R_{h,\ell}(t)) d\ell \quad (26)$$

Using the policy rules (15)–(16), total R&D labor demand can be decomposed into

$$\int_0^1 (R_{v,\ell} + R_{h,\ell}) d\ell = \underbrace{(\psi_v + \psi_h)}_{\text{fixed}} + \underbrace{\int_0^1 \mathcal{V}(w_t^H, X_t) d\ell}_{\text{variable}} \quad (27)$$

Aggregate R&D spending therefore satisfies the lower bound

$$\mathcal{E}_t \geq w_t^H (\psi_v + \psi_h) \quad (28)$$

Along the transition, productivity growth in the high-innovation sector raises labor demand and therefore the equilibrium high-skill wage, implying $\dot{w}_t^H > 0$. It follows that:

$$\dot{\mathcal{E}}_t \geq (\psi_v + \psi_h) \dot{w}_t^H > 0 \quad (29)$$

Thus, R&D expenditure rises over time despite declining innovation productivity.